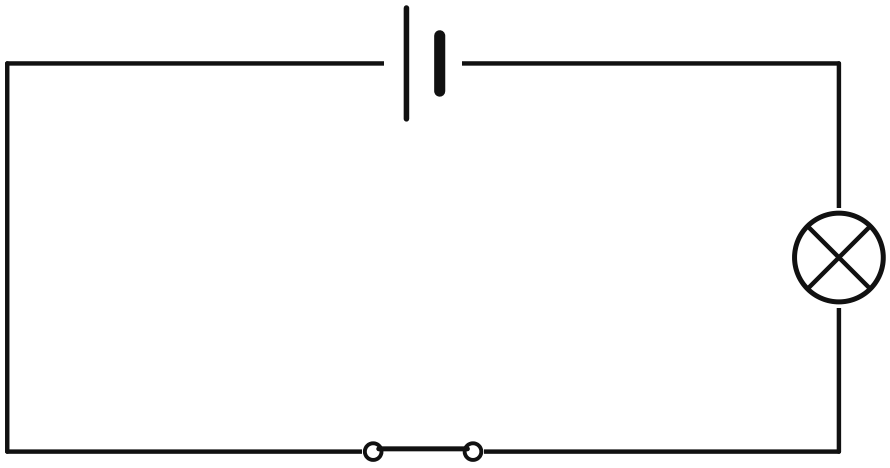
PHYSICAL AND CHEMICAL SCIENCE

Electricity and Magnets

Complete circuits, switches and breaks, a particle model of current, and magnets that attract and repel



In this pack

- **Learn it:** Circuits and magnets ★☆☆
- **Fact check:** Will it work? ★☆☆
- **Think it through:** Problem solving ★★☆
- **Investigate:** Which magnet is the strongest? ★★☆
- **Real-world case study:** Light for every home: solar power ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Circuits and magnets

Name: Class: Date:

Learning goals

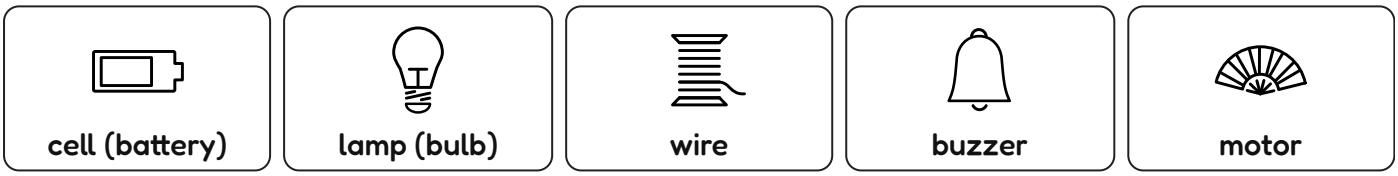
- Build a complete circuit with a cell, wires, a lamp and a switch.
- Explain why a break in a circuit stops it working, using a model of the current.
- Describe how magnets attract and repel, and which metals they attract.

Fans, lights, phones and televisions all use **electricity**. Electricity from the wall socket (mains) is very **dangerous**. In science lessons we use small **cells** (batteries), which are safe.

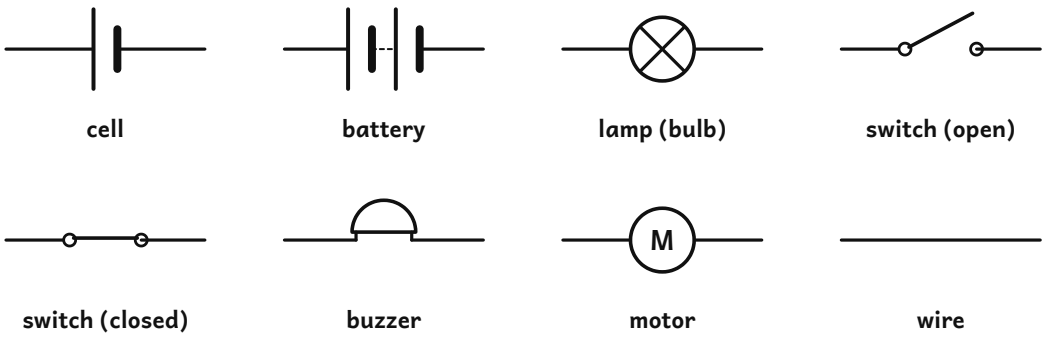
⚠ Stay safe

Never touch wall sockets, plugs or wires with wet hands. Never poke anything into a socket. Mains electricity can kill.

Parts of a circuit



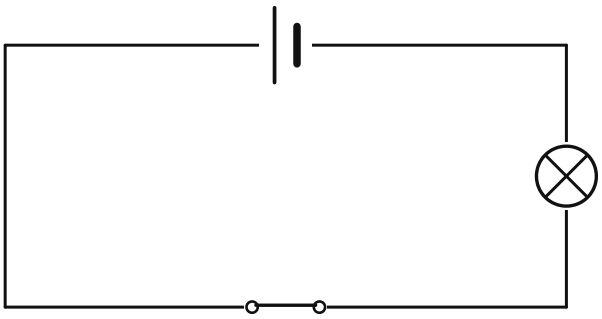
A **circuit** is a loop. Electricity flows from one end of the cell, through the wires and the lamp, and back to the other end of the cell. If the loop is **complete**, the lamp lights.



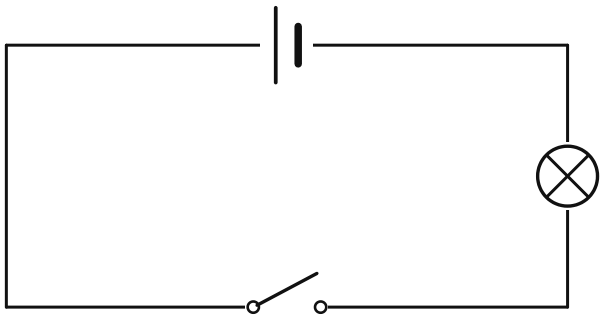
Scientists draw circuits with symbols.

Breaks in a circuit

If there is a **gap** anywhere in the loop, the electricity cannot flow and the lamp goes out. A **switch** makes a gap on purpose: switch **on** closes the gap; switch **off** opens it.



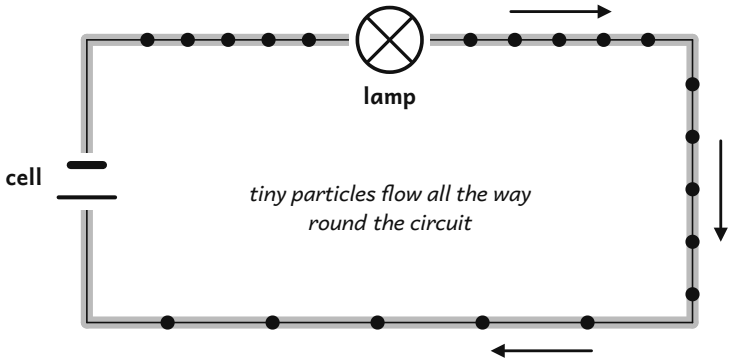
Switch on: complete circuit, lamp lights.



Switch off: a break, lamp is off.

A model of electricity

We cannot see electricity flowing. Scientists use a **model** to help: imagine tiny **particles** in the wires. The cell pushes them, and they flow **all the way round** the circuit, like children in a ring passing a ball. If the ring breaks, everything stops.

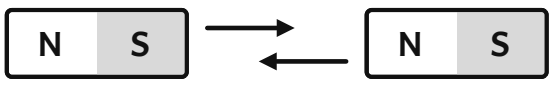


A model of an electric current.

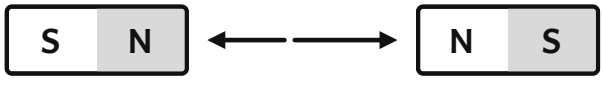
Magnets

A **magnet** pulls (**attracts**) things made of **iron** and **steel**, and a few other metals such as nickel. It does **not** attract aluminium, copper, brass, gold or silver, or non-metals like wood and plastic.

Each magnet has two ends called **poles**: **north (N)** and **south (S)**. Two different poles **attract**. Two poles that are the same **repel** (push apart).



S and N: attract (pull together)



N and N: repel (push apart)

Magnets attract and repel.

We use magnets in fridge doors, bag clasps, compasses, toys, and in big cranes that lift scrap iron.

Key words

circuit	a complete loop that electricity flows around
cell	stores electricity (often called a battery)
switch	opens or closes a gap in a circuit
attract	pull together
repel	push apart
pole	the end of a magnet, north or south

FACT CHECK · C4 · CIRCUITS AND MAGNETS

Will it work?

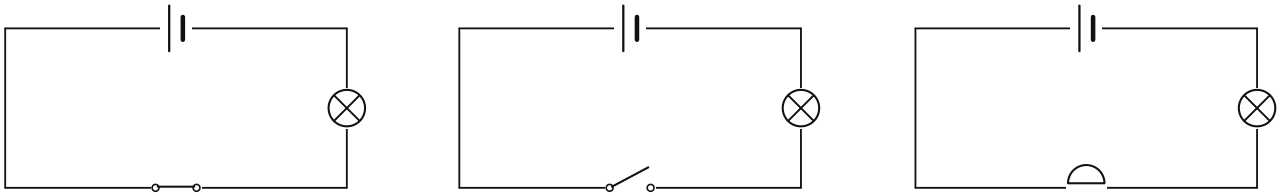
★☆☆
Grade 3

Name: **Class:** **Date:**

Learning goals

- Spot complete circuits.
- Sort magnetic metals.

1 Will the lamp light? Write yes or no. [3]



A *B* *C*

A	B	C

2 Sort the objects. [6]

SORT THESE

steel pin **iron nail** **aluminium foil** **plastic toy** **steel spoon** **copper wire**

Attracted by a magnet	Not attracted

3 Attract or repel? [3]

Poles facing	Attract or repel?
N and S	
N and N	
S and S	

Problem solving

★★☆
Grade 3

Name: Class: Date:

Learning goals

- Find faults in circuits.
- Explain with a model.

1 **Raju's torch does not work. Give three things that could be wrong.** [3]

2 **Use the particle model to explain why the lamp goes out when you open the switch.** [2]

3 **Why are electric wires covered in plastic?** [2]

INVESTIGATE · C4 · CIRCUITS AND MAGNETS

Which magnet is the strongest?

★★☆ Grade 3

Name: Class: Date:

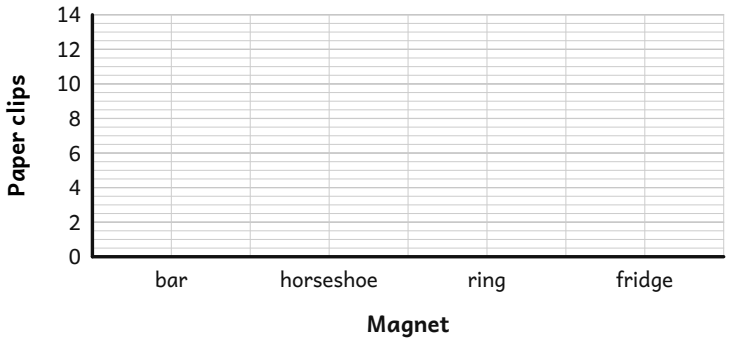
Learning goals

- Do a fair test.
- Draw a bar chart and conclude.

Class 3B hung paper clips in a chain from each magnet, one at a time, until no more would stick.

1 **Draw a bar chart.** [3]

Magnet	bar	horseshoe	ring	fridge
Paper clips	8	12	5	2



2 **Which magnet was strongest?** [1]

.....

3 **Why must they use the same kind of paper clip each time?** [2]

.....

.....

REAL-WORLD CASE STUDY · C4 · CIRCUITS AND MAGNETS

Light for every home: solar power

★★★ Grade 3

Name: **Class:** **Date:**

Learning goals

- See how a circuit changes lives.

Science | Technology | Community

Some villages and islands in Bangladesh are far from the electricity lines. Millions of families there use **solar home systems**: a **solar panel** on the roof, a **battery**, wires, **switches** and a few **LED lamps**.

In the day, the solar panel turns sunlight into electricity, which is stored in the battery. At night the battery lights the lamps, so children can study and families can charge their phones, without smoky kerosene lamps.

Based on solar home systems in rural Bangladesh.

1 **Name the parts of a solar home circuit.** [2]

2 **Why is a solar lamp better than a kerosene lamp?** [2]

3 **Draw the circuit: battery, switch and lamp.** [2]

Solar lamp circuit

Show what you know

★★★
Grade 3

Name: Class: Date:

Learning goals

- Use circuit and magnet words.

1 Find the words.

L	C	I	L	A	M	P	W	E
I	L	S	M	A	C	J	O	P
D	J	U	A	I	E	I	B	I
S	A	B	G	R	C	E	L	L
W	P	D	N	O	K	B	P	P
I	W	O	E	N	G	C	O	A
T	O	U	T	K	O	H	L	H
C	B	C	L	W	I	R	E	V
H	Y	N	H	C	C	I	K	P

cell iron lamp magnet pole switch wire

2 A lamp will light if the circuit is... [1]

◯ broken ◯ complete

◯ wet

3 Two north poles will... [1]

◯ attract ◯ repel

◯ melt

4 Challenge. How could you use a magnet to separate iron nails from sand? [2]

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
build a complete circuit	○	○	○
find a break in a circuit	○	○	○
explain current with a model	○	○	○
say when magnets attract or repel	○	○	○
name metals a magnet attracts	○	○	○